

## SECTION 1: IDENTIFICATION OF THE SUBSTANCE/MIXTURE AND OF THE COMPANY/UNDERTAKING

### Product Identifier

Product Description: Cerium(IV) oxide, nanopowder  
Cat No. : **801551**  
CAS No 1306-38-3  
Molecular Formula CeO<sub>2</sub>

### Relevant identified uses of the substance or mixture and uses advised against

Recommended Use Laboratory chemicals.  
Uses advised against No Information available

### Details of the supplier of the safety data sheet

#### **Importer**

Fisher Scientific Korea  
D5,D6, Incheon Airport Logistics Complex  
150, Gonghangdong-Ro 296 Beon-Gil  
Jung-Gu, Incheon  
Tel: +82-1661-9555  
Fax: +82-2-2023-0603

#### **Supplier**

Thermo Fisher Scientific Chemicals, Inc.  
30 Bond Street  
Ward Hill, MA 01835-8099

E-mail address Chem.KR@thermofisher.com

### Emergency Telephone Number

Emergency telephone: Medical: +(82) 070-7686-0086 or + 1-703-741-5970  
CHEMTREC: 080 822 1374 (Local), CHEMTREC: 1-800-424-9300 or + 1-703-527-3887  
Korea: 00-308-13-2549 (24 hours a day, 7 days a week)

## SECTION 2: HAZARDS IDENTIFICATION

### Classification of the substance or mixture

#### **Physical hazards**

Based on available data, the classification criteria are not met

#### **Health hazards**

Based on available data, the classification criteria are not met

#### **Environmental hazards**

Based on available data, the classification criteria are not met

### Label Elements

# SAFETY DATA SHEET

Cerium(IV) oxide, nanopowder

Revision Date 15-Feb-2024

## Other Hazards

This product does not contain any known or suspected endocrine disruptors

## NFPA

Health  
1

Flammability  
0

Instability  
0

Physical hazards  
-

## SECTION 3: COMPOSITION/INFORMATION ON INGREDIENTS

### 3.1. Substances

| Component    | Common Name    | CAS No    | Index No | Weight % |
|--------------|----------------|-----------|----------|----------|
| Cerium oxide | Cerium dioxide | 1306-38-3 | KE-05392 | 99 - 100 |

## SECTION 4: FIRST AID MEASURES

### Description of first aid measures

#### **Eye Contact**

Rinse immediately with plenty of water, also under the eyelids, for at least 15 minutes. Get medical attention.

#### **Skin Contact**

Wash off immediately with plenty of water for at least 15 minutes. Get medical attention immediately if symptoms occur.

#### **Ingestion**

Clean mouth with water and drink afterwards plenty of water. Get medical attention if symptoms occur.

#### **Inhalation**

Remove to fresh air. Get medical attention immediately if symptoms occur.

#### **Self-Protection of the First Aider**

No special precautions required.

### Most important symptoms and effects, both acute and delayed

None reasonably foreseeable.

### Indication of any immediate medical attention and special treatment needed

#### **Notes to Physician**

Treat symptomatically.

## SECTION 5: FIREFIGHTING MEASURES

### Extinguishing media

#### **Suitable Extinguishing Media**

Not combustible.

#### **Extinguishing media which must not be used for safety reasons**

No information available.

### Special hazards arising from the substance or mixture

Thermal decomposition can lead to release of irritating gases and vapors.

#### **Hazardous Combustion Products**

# SAFETY DATA SHEET

Cerium(IV) oxide, nanopowder

Revision Date 15-Feb-2024

None under normal use conditions.

## Advice for fire-fighters

As in any fire, wear self-contained breathing apparatus pressure-demand, MSHA/NIOSH (approved or equivalent) and full protective gear.

## SECTION 6: ACCIDENTAL RELEASE MEASURES

### Personal Precautions, Protective Equipment and Emergency Procedures

Ensure adequate ventilation. Use personal protective equipment as required. Avoid dust formation.

### Environmental precautions

Should not be released into the environment. See Section 12 for additional Ecological Information.

### Methods and Material for Containment and Cleaning Up

Sweep up and shovel into suitable containers for disposal. Avoid dust formation.

### Reference to Other Sections

Refer to protective measures listed in Sections 8 and 13.

## SECTION 7: HANDLING AND STORAGE

### Precautions for Safe Handling

Wear personal protective equipment/face protection. Ensure adequate ventilation. Avoid contact with skin, eyes or clothing. Avoid ingestion and inhalation. Avoid dust formation.

### Conditions for Safe Storage, Including any Incompatibilities

Keep container tightly closed in a dry and well-ventilated place.

### Specific End Uses

Use in laboratories.

## SECTION 8: EXPOSURE CONTROLS/PERSONAL PROTECTION

### Control Parameters

| Component    | CAS No    | Korea      | ACGIH TLV  | OSHA PEL   |
|--------------|-----------|------------|------------|------------|
| Cerium oxide | 1306-38-3 | Not listed | Not listed | Not listed |

| Component    | CAS No    | European Union | The United Kingdom | Germany    |
|--------------|-----------|----------------|--------------------|------------|
| Cerium oxide | 1306-38-3 | Not listed     | Not listed         | Not listed |

### ACGIH - Biological Exposure Indices

| Component    | CAS No    | ACGIH - Biological Exposure Indices |
|--------------|-----------|-------------------------------------|
| Cerium oxide | 1306-38-3 | Not listed                          |

### Exposure Controls

#### Engineering Measures

None under normal use conditions.

# SAFETY DATA SHEET

Cerium(IV) oxide, nanopowder

Revision Date 15-Feb-2024

## Personal protective equipment

|                                 |                                                    |
|---------------------------------|----------------------------------------------------|
| <b>Eye Protection</b>           | Wear safety glasses with side shields (or goggles) |
| <b>Hand Protection</b>          | Protective gloves                                  |
| <b>Skin and body protection</b> | Long sleeved clothing                              |

Inspect gloves before use.

Please observe the instructions regarding permeability and breakthrough time which are provided by the supplier of the gloves.  
(Refer to manufacturer/supplier for information)

Ensure gloves are suitable for the task: Chemical compatability, Dexterity, Operational conditions, User susceptibility, e.g. sensitisation effects, also take into consideration the specific local conditions under which the product is used, such as the danger of cuts, abrasion.

Remove gloves with care avoiding skin contamination.

|                                      |                                                                                      |
|--------------------------------------|--------------------------------------------------------------------------------------|
| <b>Personal protective equipment</b> | Use only those certified by the Korea Occupational Safety and Health Administration. |
| <b>Respiratory Protection</b>        | No protective equipment is needed under normal use conditions                        |
| <b>Recommended Filter type:</b>      | Particle filter                                                                      |

**Hygiene Measures** Handle in accordance with good industrial hygiene and safety practice

**Environmental exposure controls** No information available

## SECTION 9: PHYSICAL AND CHEMICAL PROPERTIES

### Information on basic physical and chemical properties

**Appearance (Physical State, Color, etc.)** Light yellow Solid

|                       |                          |
|-----------------------|--------------------------|
| <b>Odor</b>           | No information available |
| <b>Odor Threshold</b> | No data available        |
| <b>pH</b>             | No information available |

|                            |                          |
|----------------------------|--------------------------|
| <b>Melting Point/Range</b> | 2600 °C / 4712 °F        |
| <b>Softening Point</b>     | No data available        |
| <b>Boiling Point/Range</b> | No information available |
| <b>Flash Point</b>         | No information available |

**Method -** No information available

|                                 |                          |       |
|---------------------------------|--------------------------|-------|
| <b>Evaporation Rate</b>         | Not applicable           | Solid |
| <b>Flammability (solid,gas)</b> | No information available |       |
| <b>Explosion Limits</b>         | No data available        |       |

|                                     |                          |         |
|-------------------------------------|--------------------------|---------|
| <b>Vapor Pressure</b>               | No data available        |         |
| <b>Vapor Density</b>                | Not applicable           | Solid   |
| <b>Specific Gravity / Density</b>   | 7.132 g/cm3              | @ 20 °C |
| <b>Bulk Density</b>                 | No data available        |         |
| <b>Water Solubility</b>             | No information available |         |
| <b>Solubility in other solvents</b> | No information available |         |

### Partition Coefficient (n-octanol/water)

| Component    | CAS No    | log Pow           |
|--------------|-----------|-------------------|
| Cerium oxide | 1306-38-3 | No data available |

# SAFETY DATA SHEET

Cerium(IV) oxide, nanopowder

Revision Date 15-Feb-2024

Autoignition Temperature No data available  
Decomposition Temperature No data available  
Viscosity Not applicable Solid  
Explosive Properties No information available  
Oxidizing Properties No information available

Molecular Formula CeO<sub>2</sub>  
Molecular Weight 172.12

## SECTION 10: STABILITY AND REACTIVITY

Reactivity None known, based on information available

Chemical Stability Stable under normal conditions.

### Possibility of Hazardous Reactions

Hazardous Polymerization No information available.  
Hazardous Reactions None under normal processing.

Conditions to Avoid None known.

Incompatible Materials Oxidizing agent.

Hazardous Decomposition Products None under normal use conditions.

## SECTION 11: TOXICOLOGICAL INFORMATION

### 11.1. Information on hazard classes as defined in Regulation (EC) No 1272/2008

#### Product Information

#### Information on expected route of exposure

Inhalation Not an expected route of exposure.  
Ingestion No known effect based on information supplied.  
Eyes Not an expected route of exposure.  
Skin No known effect based on information supplied.

#### Information on Health Hazards

##### (a) acute toxicity;

Oral Based on available data, the classification criteria are not met  
Dermal Based on available data, the classification criteria are not met  
Inhalation Based on available data, the classification criteria are not met

| Component    | CAS No    | LD50 Oral             | LD50 Dermal               | LC50 Inhalation              |
|--------------|-----------|-----------------------|---------------------------|------------------------------|
| Cerium oxide | 1306-38-3 | LD50 > 5 g/kg ( Rat ) | LD50 > 2000 mg/kg ( Rat ) | LC50 > 5.05 mg/L ( Rat ) 4 h |

# SAFETY DATA SHEET

Cerium(IV) oxide, nanopowder

Revision Date 15-Feb-2024

(b) skin corrosion/irritation; No data available

(c) serious eye damage/irritation; No data available

(d) respiratory or skin sensitization;  
Respiratory No data available  
Skin No data available

| Component    | CAS No    | Test method       | Test species      | Study result      |
|--------------|-----------|-------------------|-------------------|-------------------|
| Cerium oxide | 1306-38-3 | No data available | No data available | No data available |

(e) germ cell mutagenicity; No data available

| Component    | CAS No    | Test method       | Test species      | Study result      |
|--------------|-----------|-------------------|-------------------|-------------------|
| Cerium oxide | 1306-38-3 | No data available | No data available | No data available |

(f) carcinogenicity; No data available

| Component    | CAS No    | Test method       | Test species / Duration | Study result      |
|--------------|-----------|-------------------|-------------------------|-------------------|
| Cerium oxide | 1306-38-3 | No data available | No data available       | No data available |

There are no known carcinogenic chemicals in this product

| Component    | CAS No    | IARC       | NTP        | ACGIH      | OSHA       | UK         |
|--------------|-----------|------------|------------|------------|------------|------------|
| Cerium oxide | 1306-38-3 | Not listed | Not listed | Not listed | Not listed | Not listed |

(g) reproductive toxicity; No data available

| Component    | CAS No    | Test method       | Test species / Duration | Study result      |
|--------------|-----------|-------------------|-------------------------|-------------------|
| Cerium oxide | 1306-38-3 | No data available | No data available       | No data available |

(h) STOT-single exposure; No data available

(i) STOT-repeated exposure; No data available

Target Organs No information available.

(j) aspiration hazard; Not applicable  
Solid

Other Adverse Effects  
No information available.

| Component    | CAS No    | EU - Endocrine Disruptors Candidate List | EU - Endocrine Disruptors - Evaluated Substances | Japan - Endocrine Disruptor Information |
|--------------|-----------|------------------------------------------|--------------------------------------------------|-----------------------------------------|
| Cerium oxide | 1306-38-3 | Not applicable                           | Not applicable                                   | Not applicable                          |

## SECTION 12: ECOLOGICAL INFORMATION

Ecotoxicity effects

| Component | CAS No | Freshwater Fish | Water Flea | Freshwater Algae | Microtox |
|-----------|--------|-----------------|------------|------------------|----------|
|-----------|--------|-----------------|------------|------------------|----------|

ALFAA801551

# SAFETY DATA SHEET

Cerium(IV) oxide, nanopowder

Revision Date 15-Feb-2024

|              |           |                   |                   |                   |                   |
|--------------|-----------|-------------------|-------------------|-------------------|-------------------|
| Cerium oxide | 1306-38-3 | No data available | No data available | No data available | No data available |
|--------------|-----------|-------------------|-------------------|-------------------|-------------------|

**Persistence and degradability**  
**Degradability** No information available  
Not relevant for inorganic substances.

**Bioaccumulative potential** No information available

**Mobility in soil** No information available

**Ozone Depletion Potential**

| Component    | CAS No    | Ozone Depletion Potential |
|--------------|-----------|---------------------------|
| Cerium oxide | 1306-38-3 | Not listed                |

**Other adverse effects** No information available

## SECTION 13: DISPOSAL CONSIDERATIONS

**Waste treatment methods**

**Waste from Residues/Unused Products** Chemical waste generators must determine whether a discarded chemical is classified as a hazardous waste. Chemical waste generators must also consult local, regional, and national hazardous waste regulations to ensure complete and accurate classification.

**Contaminated Packaging** Empty containers should be taken to an approved waste handling site for recycling or disposal.

## SECTION 14: TRANSPORT INFORMATION

**Road and Rail Transport** Not Regulated

**IATA** Not regulated

**IMDG/IMO** Not regulated  
**Marine Pollutant** No hazards identified

**Special Precautions for User** No special precautions required

## SECTION 15: REGULATORY INFORMATION

**Safety, health and environmental regulations/legislation specific for the substance or mixture**

Legend: X - Listed '-' - Not Listed

**International Inventories**

| Component    | CAS No    | KECL     | TSCA | EINECS    | IECSC | DSL | NDSL | PICCS | ENCS | ISHL | AICS |
|--------------|-----------|----------|------|-----------|-------|-----|------|-------|------|------|------|
| Cerium oxide | 1306-38-3 | KE-05392 | X    | 215-150-4 | X     | X   | -    | X     | X    | X    | X    |

| Component    | CAS No    | Seveso III Directive (2012/18/EC) - Qualifying Quantities for Major Accident Notification | Seveso III Directive (2012/18/EC) - Qualifying Quantities for Safety Report Requirements | Rotterdam Convention (PIC) | Basel Convention (Hazardous Waste) |
|--------------|-----------|-------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------|----------------------------|------------------------------------|
| Cerium oxide | 1306-38-3 | Not applicable                                                                            | Not applicable                                                                           | Not applicable             | Not applicable                     |

| Component | CAS No | OECD HPV | Persistent Organic | Ozone Depletion |
|-----------|--------|----------|--------------------|-----------------|
|-----------|--------|----------|--------------------|-----------------|

ALFAA801551

# SAFETY DATA SHEET

Cerium(IV) oxide, nanopowder

Revision Date 15-Feb-2024

|              |           |        | Pollutant      | Potential      |
|--------------|-----------|--------|----------------|----------------|
| Cerium oxide | 1306-38-3 | Listed | Not applicable | Not applicable |

## Korean National Regulations

| Component    | CAS No    | Act on Registration and Evaluation of Chemical Substances (K-REACH) | Authorised Chemicals | Existing Substances Subject to Registration |
|--------------|-----------|---------------------------------------------------------------------|----------------------|---------------------------------------------|
| Cerium oxide | 1306-38-3 | Annex 1 - KE-05392                                                  | Not applicable       | Not applicable                              |

| Component    | CAS No    | Chemical Control Act - Toxic Chemicals | Chemical Control Act - Prohibited Chemicals | Chemical Control Act - Use Restricted Chemicals |
|--------------|-----------|----------------------------------------|---------------------------------------------|-------------------------------------------------|
| Cerium oxide | 1306-38-3 | Not applicable                         | Not applicable                              | Not applicable                                  |

| Component    | CAS No    | Chemical Control Act - Accident Precaution Chemicals (% in mixtures) | Chemical Control Act - Accident Precaution Chemicals - Quantity limits Storage (% in mixtures) | Chemical Control Act - Accident Precaution Chemicals - Quantity limits Manufacture/Use (% in mixtures) |
|--------------|-----------|----------------------------------------------------------------------|------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------|
| Cerium oxide | 1306-38-3 | Not applicable                                                       | Not applicable                                                                                 | Not applicable                                                                                         |

| Component    | CAS No    | Waste Control Law | Ministry of Environment - CMR risk | Ministry of Environment - Critically Controlled Substance |
|--------------|-----------|-------------------|------------------------------------|-----------------------------------------------------------|
| Cerium oxide | 1306-38-3 | Not applicable    | Not applicable                     | Not applicable                                            |

| Component    | CAS No    | ISHA - Harmful Agents Subject to Work Environment Monitoring | ISHA - Prohibited substances | ISHA - Substances requiring permission |
|--------------|-----------|--------------------------------------------------------------|------------------------------|----------------------------------------|
| Cerium oxide | 1306-38-3 | Not applicable                                               | Not applicable               | Not applicable                         |

| Component    | CAS No    | ISHA - Substances subject to control | ISHA - Harmful Agents Requiring Health Examination | ISHA - Permissible Exposure Limits |
|--------------|-----------|--------------------------------------|----------------------------------------------------|------------------------------------|
| Cerium oxide | 1306-38-3 | Not applicable                       | Not applicable                                     | Not applicable                     |

| Component    | CAS No    | ISHA - Subject to Process Safety Reports (minimum quantity) | ISHA - Threshold Limit Values (TLVs) Chemicals | ISHA - Special management materials |
|--------------|-----------|-------------------------------------------------------------|------------------------------------------------|-------------------------------------|
| Cerium oxide | 1306-38-3 | Not applicable                                              | Not applicable                                 | Not applicable                      |

## National Fire Association - Dangerous Substances Minimum quantity requiring a permit

| Component    | CAS No    | Class 1 - Oxidising solids | Class 2 - Flammable solid | Class 3 - Spontaneously Combustible Substances and Dangerous Substances When Wet | Class 4 - Flammable liquids | Class 5 - Self-reactive substances | Class 6 - Oxidising liquids |
|--------------|-----------|----------------------------|---------------------------|----------------------------------------------------------------------------------|-----------------------------|------------------------------------|-----------------------------|
| Cerium oxide | 1306-38-3 | Not applicable             | Not applicable            | Not applicable                                                                   | Not applicable              | Not applicable                     | Not applicable              |

## Control Parameters

| Component    | CAS No    | Korea      | ACGIH - Biological Exposure Indices |
|--------------|-----------|------------|-------------------------------------|
| Cerium oxide | 1306-38-3 | Not listed | Not listed                          |

## US Management Information

OSHA - Occupational Safety and Health Administration

ALFAA801551



# SAFETY DATA SHEET

Cerium(IV) oxide, nanopowder

Revision Date 15-Feb-2024

Not applicable

| Component    | CAS No    | Specifically Regulated Chemicals | Highly Hazardous Chemicals |
|--------------|-----------|----------------------------------|----------------------------|
| Cerium oxide | 1306-38-3 | Not applicable                   | Not applicable             |

CERCLA

Not applicable

| Component    | CAS No    | CERCLA Extremely Hazardous Substances RQs | Hazardous Substances RQs | SARA 313 - Threshold Values % |
|--------------|-----------|-------------------------------------------|--------------------------|-------------------------------|
| Cerium oxide | 1306-38-3 | Not applicable                            | Not applicable           | Not applicable                |

## CLP Classification - According to GB-CLP Regulations UK SI 2019/720 and UK SI 2020/1567

Based on available data, the classification criteria are not met.

## SECTION 16: OTHER INFORMATION

### Legend

**CAS** - Chemical Abstracts Service

**EINECS/ELINCS** - European Inventory of Existing Commercial Chemical Substances/EU List of Notified Chemical Substances

**PICCS** - Philippines Inventory of Chemicals and Chemical Substances

**IECSC** - Chinese Inventory of Existing Chemical Substances

**KECL** - Korean Existing and Evaluated Chemical Substances

**TSCA** - United States Toxic Substances Control Act Section 8(b) Inventory

**DSL/NDL** - Canadian Domestic Substances List/Non-Domestic Substances List

**ENCS** - Japanese Existing and New Chemical Substances

**AICS** - Australian Inventory of Chemical Substances

**NZIoC** - New Zealand Inventory of Chemicals

**WEL** - Workplace Exposure Limit

**ACGIH** - American Conference of Governmental Industrial Hygienists

**RPE** - Respiratory Protective Equipment

**LC50** - Lethal Concentration 50%

**POW** - Partition coefficient Octanol:Water

**TWA** - Time Weighted Average

**IARC** - International Agency for Research on Cancer

**LD50** - Lethal Dose 50%

**EC50** - Effective Concentration 50%

**ADR** - European Agreement Concerning the International Carriage of Dangerous Goods by Road

**IMO/IMDG** - International Maritime Organization/International Maritime Dangerous Goods Code

**OECD** - Organisation for Economic Co-operation and Development

**BCF** - Bioconcentration factor

**ICAO/IATA** - International Civil Aviation Organization/International Air Transport Association

**MARPOL** - International Convention for the Prevention of Pollution from Ships

**ATE** - Acute Toxicity Estimate

**VOC** - (Volatile Organic Compound)

### Key literature references and sources for data

<https://echa.europa.eu/information-on-chemicals>

Suppliers safety data sheet, Chemadvisor - LOLI, Merck index, RTECS

### Training Advice

Chemical hazard awareness training, incorporating labelling, Safety Data Sheets (SDS), Personal Protective Equipment (PPE) and hygiene.

### Prepared By

Health, Safety and Environmental Department

### Creation Date

15-Apr-2015

### Revision Date

15-Feb-2024

### Revision Number

1

### Revision Summary

Initial Release.

MOEL's Public Notice No. 2023-9 (Standards for Classification and Labeling of Chemical

# SAFETY DATA SHEET

Cerium(IV) oxide, nanopowder

Revision Date 15-Feb-2024

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## Substances and Safety Data Sheets)

### Disclaimer

The information provided in this Safety Data Sheet is correct to the best of our knowledge, information and belief at the date of its publication. The information given is designed only as a guidance for safe handling, use, processing, storage, transportation, disposal and release and is not to be considered a warranty or quality specification. The information relates only to the specific material designated and may not be valid for such material used in combination with any other materials or in any process, unless specified in the text

**End of Safety Data Sheet**